

Escenario: CaribeTech — Nueva sucursal

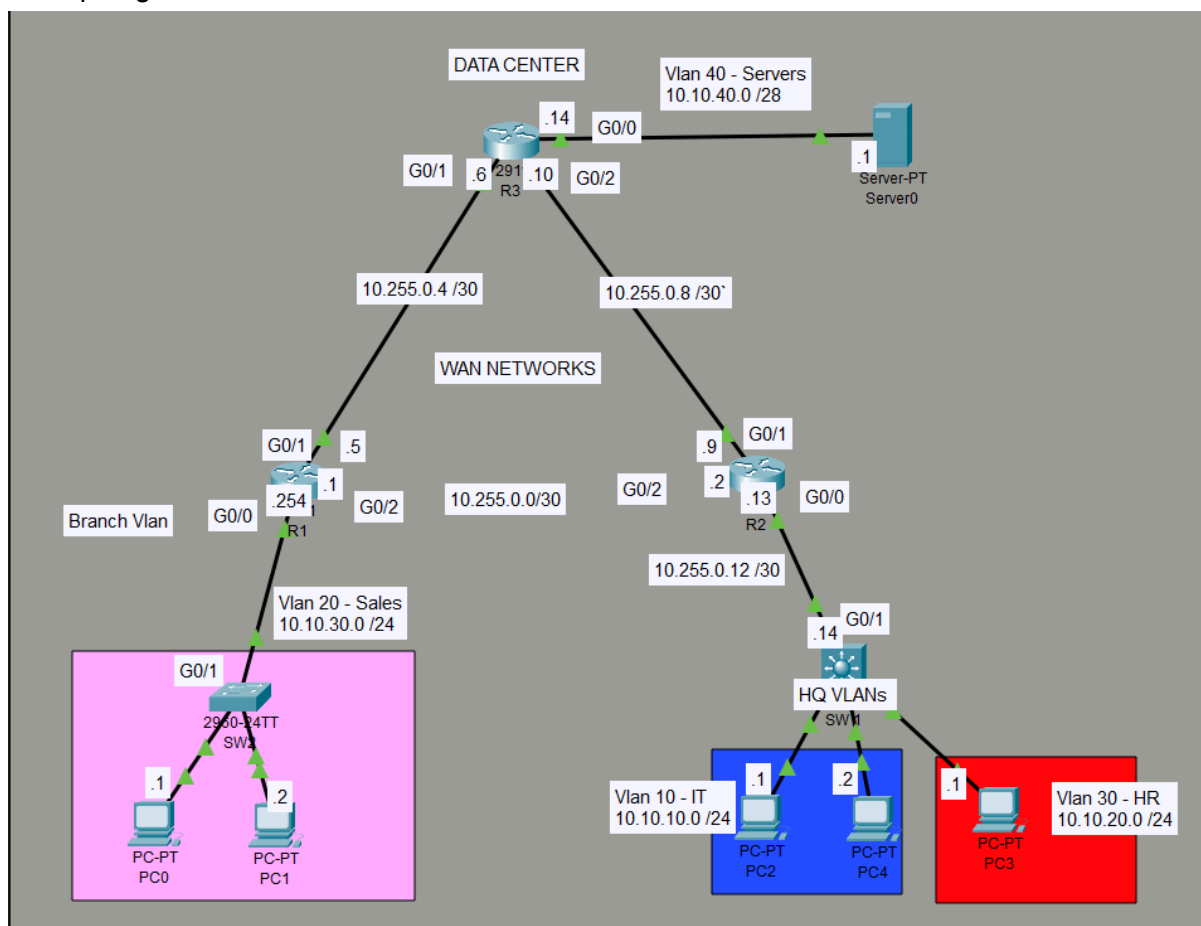
CaribeTech tiene ahora **tres routers** que conectan:

- Headquarters
- Branch Office
- Data Center

La empresa quiere dejar de mantener rutas estáticas manualmente y utilizar **OSPF**.

El objetivo de esta laboratorio es construir una red **OSPF Area 0** completamente funcional.

Topología



Objetivos

Parte 1 — Switching

En **SW1 — 3560 L3**:

Configura:

- **VLAN 10**
 - vlan 10: Create the vlan 10
 - name IT: Name the vlan
- **VLAN 20**
 - vlan 20: Create the vlan 20
 - name sales: Name the vlan
- **SVIs**
 - Vlan 10
 - Interface Vlan 10: Create vlan interface
 - ip add 10.10.10.254 255.255.255.0: Assign the IP to the interface
 - Vlan 20
 - Interface Vlan 20: Create vlan interface
 - ip add 10.10.20.254 255.255.255.0: Assign the IP to the interface
- **Inter-VLAN routing**
 - ip routing
 - Interface f0/1: Enter port FastEthernet 0/1
 - switchport mode acc: Change port to access mode
 - switchport acc vlan 10: Assign port to the vlan 10.
 - Interface f0/2: Enter port FastEthernet 0/2
 - switchport mode acc: Change port to access mode
 - switchport acc vlan 20: Assign port to the vlan 20.

En **SW2 — 2960**:

Configura:

- **VLAN 30**
 - Vlan 30: Creates vlan 30
 - Name HR: Assign a name to the vlan
- **Access ports**
 - Interface range f0/1-2: Enter ports FastEthernet 0/1 and 0/2
 - switchport mode acc: Change ports to access mode
 - switchport acc vlan 30: Assign port to the vlan 30.
- **Uplink hacia R2**
 - Interface g0/0: Enter interface GigabitEthernet 0/0

- switchport mode trunk: Change the port to a trunking state
- switchport trunk allowed vlan 30: Allow traffic tag with vlan 30 to pass through.

Parte 2 — Routing

On Switch 1:

- ip routing: Put the switch on a layer 3 mode
- int g 0/0: Enters the interfaces
- no switchport: Changes the switch interface to a Layer 3
- ip add 10.255.0.14 255.255.255.252: Assigning IP address to the interface
- no sh: turn the interface up.

On R1:

- int g 0/0: Enters the interfaces
- ip add 10.10.30.254 255.255.255.252: Assigning IP address to the interface
- no sh: turn the interface up.

Repeat the process until all needed interfaces have an assigned IP address.

Parte 3 — OSPF

Configura **OSPF single-area**.

Todos los routers deben participar en: Area 0

Debes anunciar:

- Redes de HQ:
 - On Router 2:
 - router ospf 1: Enables ospf
 - Network 10.10.10. 0.0.0.255(/24) area 0: Announced the network
 - Network 10.10.30.0 0.0.0.255(/24) area 0: Announced the network
- Redes de Branch
 - On Switch 1:
 - router ospf 1: Enables ospf
 - Network 10.10.20.0. 0.0.0.255(/24) area 0: Announced the network
 - Network 10.10.10.0.0. 0.0.0.255(/24) area 0: Announced the network
- Red del Data Center:
 - On Router 3:
 - router ospf 1: Enables ospf
 - Network 10.10.40. 0.0.0.255(/24) area 0: Announced the network
- Redes WAN
 - On Router 1:
 - router ospf 1: Enables ospf
 - Network 10.255.0.0 0.0.0.3(/30) area 0: Announced the network
 - Network 10.255.0.4 0.0.0.3(/30) area 0: Announced the network
 - On Router 2:
 - router ospf 1: Enables ospf
 - Network 10.255.0.0 0.0.0.3(/30) area 0: Announced the network
 - Network 10.255.0.8 0.0.0.3(/30) area 0: Announced the network
 - Network 10.255.0.12 0.0.0.3(/30) area 0: Announced the network
 - On Router 3:
 - router ospf 1: Enables ospf
 - Network 10.255.0.4 0.0.0.3(/30) area 0: Announced the network
 - Network 10.255.0.8 0.0.0.3(/30) area 0: Announced the network
 - On Switch 1:
 - router ospf 1: Enables ospf
 - Network 10.255.0.12 0.0.0.3(/30) area 0: Announced the network

Note: This configuration needs to be done on every router and the Layer 3 Switch

OSPF Challenge #1

Configurar **Router IDs** manualmente:

On Router 1:

- router-id 1.1.1.1

On Router 2:

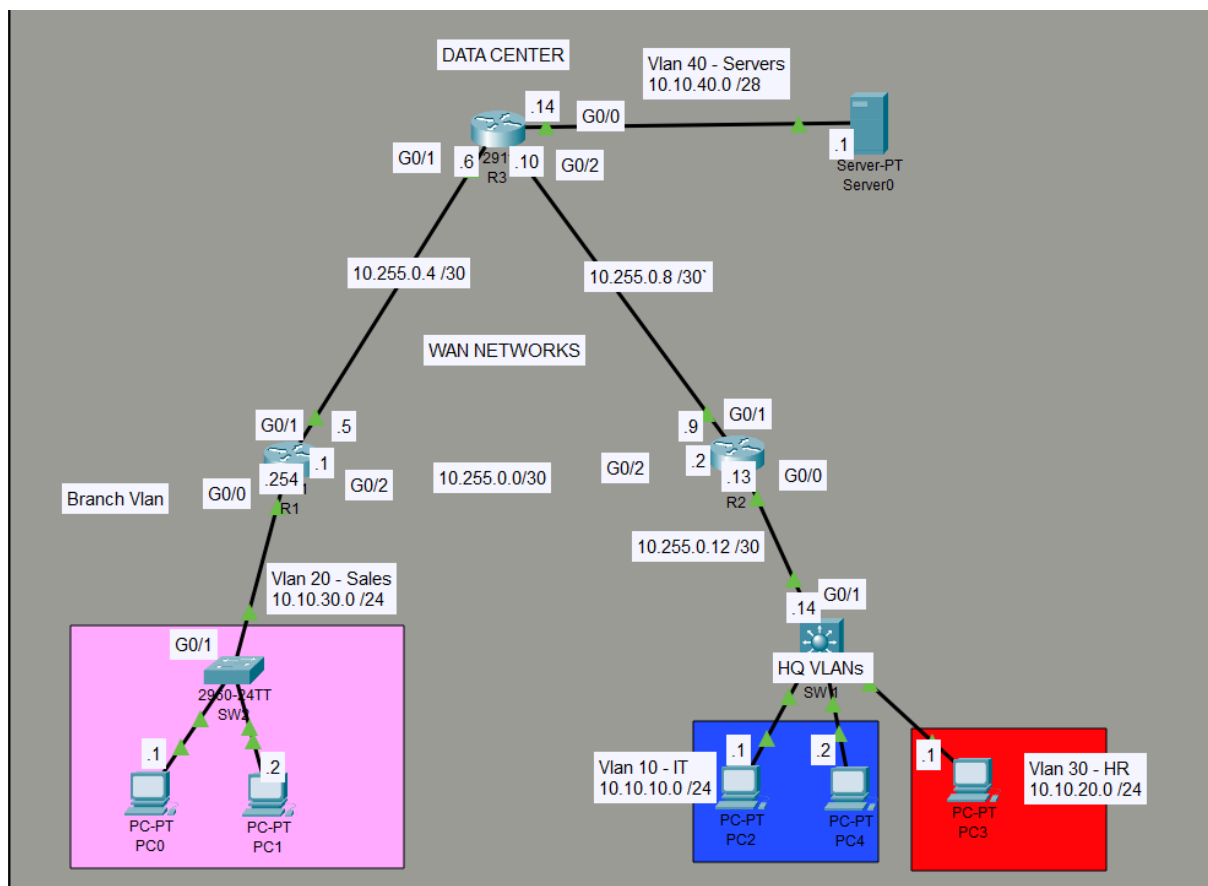
- router-id 2.2.2.2

And like that until all routers participating on OSPF have a router id.

OSPF Troubleshooting Challenge

Ahora viene una parte importante.

Una vez que todo funcione, **desconecta físicamente el enlace: R1 - R2**



Después revisa:

show ip route

```
down of detached
sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 9 subnets, 4 masks
O       10.10.10.0/24 [110/2] via 10.255.0.14, 05:53:10, GigabitEthernet0/0
O       10.10.20.0/24 [110/2] via 10.255.0.14, 05:53:10, GigabitEthernet0/0
O       10.10.30.0/24 [110/3] via 10.255.0.10, 00:01:01, GigabitEthernet0/1
O       10.10.40.0/28 [110/2] via 10.255.0.10, 06:24:40, GigabitEthernet0/1
O       10.255.0.4/30 [110/2] via 10.255.0.10, 00:01:01, GigabitEthernet0/1
C       10.255.0.8/30 is directly connected, GigabitEthernet0/1
L       10.255.0.9/32 is directly connected, GigabitEthernet0/1
C       10.255.0.12/30 is directly connected, GigabitEthernet0/0
L       10.255.0.13/32 is directly connected, GigabitEthernet0/0

R2#
```

show ip ospf neighbor

```
R2#sh ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
4.4.4.4        1     FULL/DR         00:00:30    10.255.0.14    GigabitEthernet0/0
3.3.3.3        1     FULL/DR         00:00:39    10.255.0.10    GigabitEthernet0/1
R2#
```

OSPF Challenge #2 — Cost

Ahora quiero que experimentes con **OSPF cost**.

Haz que el enlace: R1 ↔ R3 sea menos preferido que: R1 ↔ R2

- int g0/1: Enters the interface
- ip ospf cost 10: Give the cost to the route

Después utiliza:

show ip route

```

R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 11 subnets, 4 masks
O       10.10.10.0/24 [110/2] via 10.255.0.14, 05:58:50, GigabitEthernet0/0
O       10.10.20.0/24 [110/2] via 10.255.0.14, 05:58:50, GigabitEthernet0/0
O       10.10.30.0/24 [110/2] via 10.255.0.1, 00:00:19, GigabitEthernet0/2
O       10.10.40.0/28 [110/3] via 10.255.0.1, 00:00:19, GigabitEthernet0/2
C       10.255.0.0/30 is directly connected, GigabitEthernet0/2
L       10.255.0.2/32 is directly connected, GigabitEthernet0/2
O       10.255.0.4/30 [110/2] via 10.255.0.1, 00:00:19, GigabitEthernet0/2
C       10.255.0.8/30 is directly connected, GigabitEthernet0/1
L       10.255.0.9/32 is directly connected, GigabitEthernet0/1
C       10.255.0.12/30 is directly connected, GigabitEthernet0/0
L       10.255.0.13/32 is directly connected, GigabitEthernet0/0
==More==

```

y:

traceroute

```

R2#traceroute 10.10.30.1
Type escape sequence to abort.
Tracing the route to 10.10.30.1

 1  10.255.0.1          0 msec      0 msec      0 msec
 2  10.10.30.1          24 msec     0 msec      0 msec

```

para comprobar qué camino selecciona OSPF.

Luego rompe el enlace preferido y observa qué sucede.

```

traceroute 10.10.30.1
Type escape sequence to abort.
Tracing the route to 10.10.30.1

 1  10.255.0.10         0 msec      0 msec      0 msec
 2  10.255.0.5          0 msec      0 msec      0 msec
 3  10.10.30.1          0 msec      0 msec      0 msec
R2#

```

Additional commands:

- Passive-interface G0/0: Make the interfaces passive and do not try to make unnecessary ospf neighbors.
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